

GRADUATION THESIS

A MULTI-AGENT SYSTEM FOR MOVIE RECOMMENDATION

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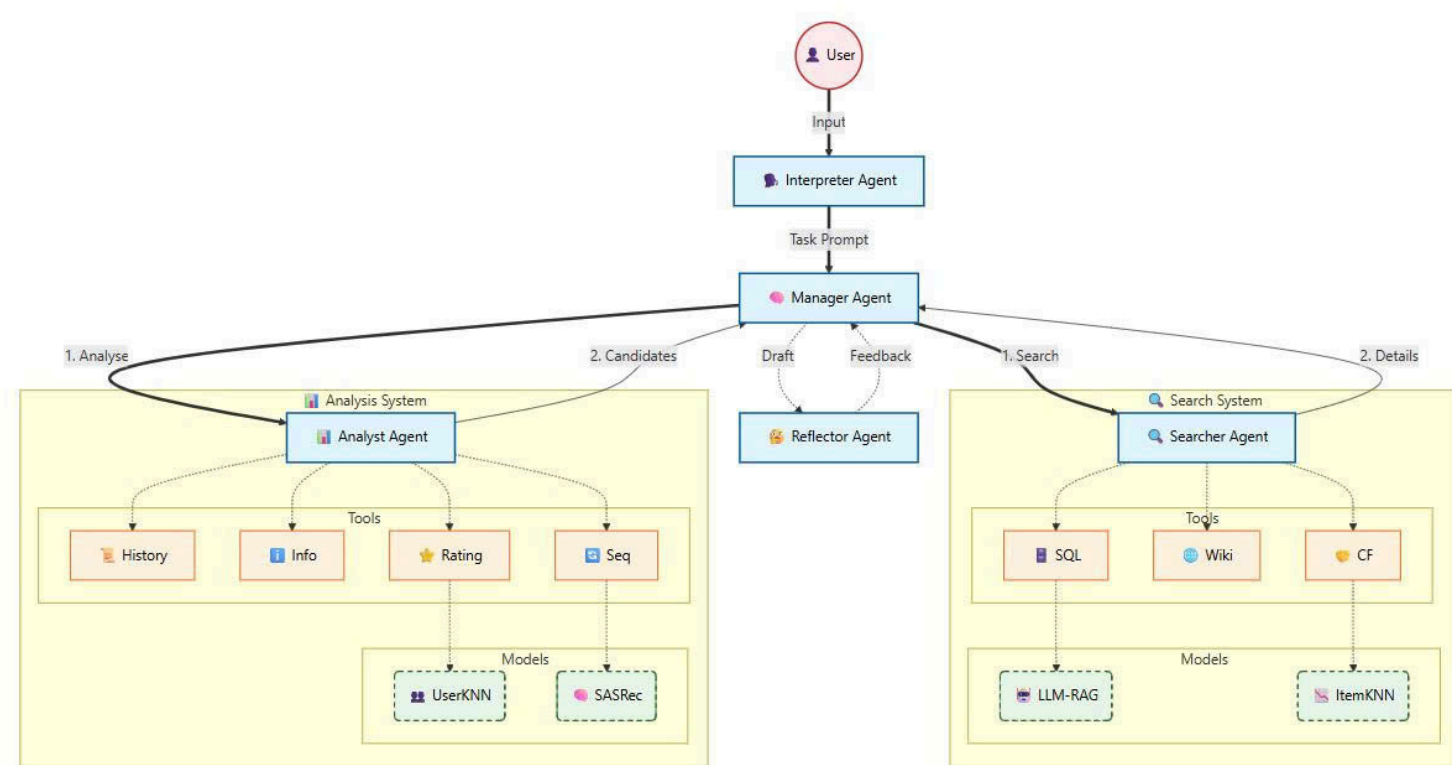
OVERVIEW

- Traditional recommender systems suffer from limitations in interpretability, scalability, and flexibility.
- Recent advances in large language models (LLMs) and multi-agent techniques enable more explicit and structured modeling of user behavior.
- Employing a multi-agent framework for movie recommendation enhances the personalization of user experiences.

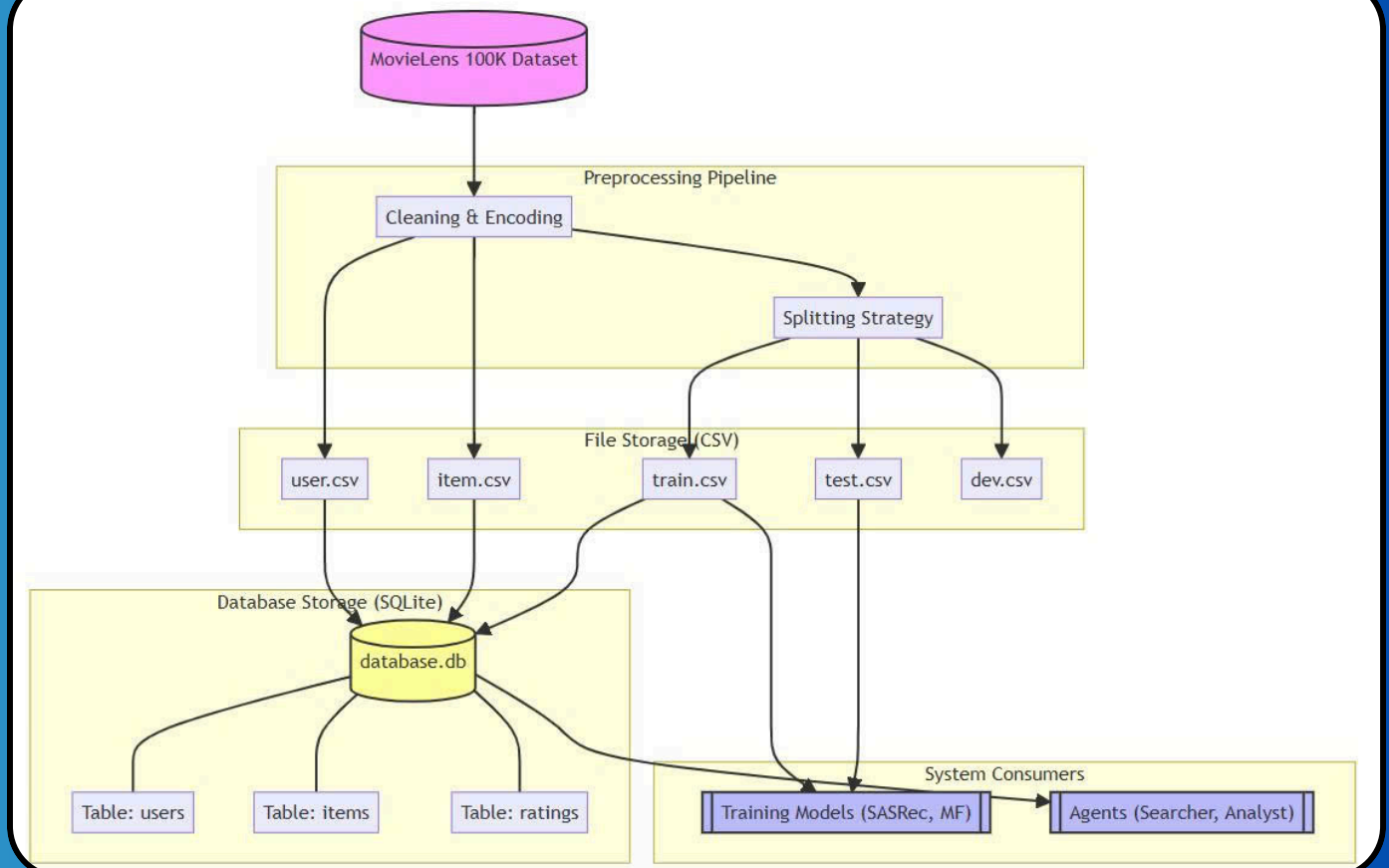
OBJECTIVE

- Designs and implements a movie recommendation system based on a Multi-Agent System to improve personalization and enhance recommendation quality.
- Each agent is assigned a specialized task to increase system flexibility.
- The proposed system is expected to serve as a foundational framework for large-scale real-world recommender applications.

SYSTEM ARCHITECTURE



DATA FLOW AND STORAGE



RESULTS

TECHNOLOGIES



Suggest 5 movies for user ID 123

System Process

- Interpreter: Finish with results:
 - Hãy dự đoán danh sách 5 bộ phim cho người dùng có ID 123.
- Manager: Thought 1: The user has requested movie recommendations based on their preferences. Since I need to analyze the user's preferences first, I will proceed with the appropriate action.
- Manager: Calling Analyst with user, 123.
- Manager: Response from Analyst with user, 123:
 - User 123 has a diverse interaction history with a total of 20 items, showcasing a range of preferences. The user has rated several items highly, with notable ratings of 5.0 for items 432, 5, 205, and 4.0 for items 185, 147, 275, and 276. This indicates a strong preference for high-quality content, particularly in genres represented by these items. However, the user also displayed some mixed ratings, such as a 1.0 for item 255 and 3.0 for items 50 and 121. This suggests that while the user enjoys certain aspects of the items they rated highly, they may have specific expectations or preferences that were not met by the lower-rated items. For instance, item 255 could have lacked engaging storytelling or character development, which are crucial for the user's enjoyment.
 - Overall, user 123 seems to favor well-crafted narratives and engaging content, likely in genres such as Action and Drama, as inferred from their high ratings. This analysis can guide future recommendations by focusing on similar high-quality items that align with their demonstrated preferences.
- Manager: Thought 2: Action 1: Search[Requirements]
- Manager: CF Recommendation for User 123.
- CF Recommendations for User 123 (top 10): Title: Fast Five (2015), Genres: Action (score: 4.8), 1512: Title: World of Apo, The (1996), Genres: Drama (score: 4.8), 1390: Title: Star Wars (1977), Genres: Drama (score: 4.8), 1445: Title: Father of the Bride (1991), Genres: Drama (score: 4.5), 1450: Title: Baywatch (1989), Genres: Drama (score: 4.4), 1388: Title: G.I. Joe: The Movie (1986), Genres: Drama (score: 4.3), 272: Title: Good Will Hunting (1997), Genres: Drama (score: 4.3), 251: Title: Shall We Dance? (1996), Genres: Comedy (score: 4.3), 1553: Title: Gold Diggers: The Secret of Bear Mountain (1935), Genres: Adventure (score: 4.3), 109: Title: Mirror Trossen, The (1945), Genres: Animation (score: 4.2).

1. USER PREFERENCE SUMMARY: Based on the analysis, User 123 enjoys a mix of genres including Comedy, Action, and Drama, indicating a preference for engaging and entertaining narratives.

2. 5 MOVIE RECOMMENDATIONS:

1. Mighty Aphrodite (1995) - Comedy
 - Summary: A sportswriter discovers that his adopted son is a gifted athlete and sets out to find his biological mother, leading to unexpected comedic situations.
 - Reason: This film aligns with your taste for comedies that blend humor with heartfelt moments.
2. Star Wars (1977) - Action/Adventure/Romance/Sci-Fi/War
 - Summary: A young farm boy joins forces with a band of rebels to fight against the tyrannical Galactic Empire and rescue Princess Leia.
 - Reason: As a classic in the action and adventure genre, this film promises an epic journey that matches your interest in thrilling narratives.
3. Fargo (1996) - Crime/Drama/Thriller
 - Summary: A pregnant police chief investigates a series of homicides that ensue from a kidnapping gone wrong in the snowy Midwest.
 - Reason: This film's unique blend of dark humor and crime drama fits well with your appreciation for complex storytelling.
4. Raiders of the Lost Ark (1981) - Action/Adventure
 - Summary: Archaeologist Indiana Jones races against time to find the biblical Ark of the Covenant before the Nazis can harness its powers.
 - Reason: This action-packed adventure is perfect for your love of thrilling escapades and historical lore.
5. Broken Arrow (1996) - Action/Thriller
 - Summary: A military pilot must stop a rogue general from using stolen nuclear weapons against the United States.
 - Reason: This film's high-stakes action and suspense align with your preference for adrenaline-pumping thrillers.

3. SELECTION CRITERIA: These five movies were selected based on their alignment with your viewing history, ensuring a mix of genres that you enjoy while also introducing you to some classics.

4. FRIENDLY CLOSING: Enjoy your movie watching! If you have any specific genres or themes in mind for future recommendations, feel free to ask!

Predict rating of User 123 for movie 456

System Process

- Interpreter: Finish with results:
 - Hãy dự đoán đánh giá của người dùng 123 cho bộ phim 456.
- Manager: Thought 1: I need to predict the rating for User 123 for movie 456. Since I have both the user ID and item ID, I will proceed with the analysis step immediately.
- Manager: Calling Analyst with user, 123, item, 456.
- Manager: Response from Analyst with user, 123, item, 456:
 - User 123 has a diverse interaction history with a mix of ratings. The ratings for the items are as follows: 3.0, 5.0, 4.0, 5.0, 3.5, 4.0, 5.0, 4.0, 1.0, and 3.0. This indicates that the user has strong preferences for certain items, particularly those rated 4.0 and above, which suggest a liking for high-quality content.
 - The user has rated several items highly (5.0 ratings for items 432, 5, 275, and 205), indicating a strong preference for the features of these items. However, the presence of lower ratings (3.0 and 1.0) suggests that the user is discerning and does not hesitate to express dissatisfaction with items that do not meet their expectations.
 - The high ratings may correlate with specific genres or themes that resonate with the user, while the lower ratings could indicate a mismatch in genre or quality. For instance, if the highly rated items are predominantly in the Action or Sci-Fi genres, it would suggest that these are the types of content that the user enjoys most. Conversely, the lower rated items may belong to genres or styles that the user finds less appealing.
 - Overall, User 123 appears to be a selective viewer who appreciates quality content, particularly in genres that align with their tastes. This analysis can guide future recommendations to focus on high-rated genres or similar items to those that have previously garnered high ratings.
- Manager: Thought 2: Action 2: Finish[USER ANALYSIS]. User 123 has a diverse interaction history with a mix of ratings. They have rated several items highly, particularly enjoying Fantasia, Dead Man Walking, Sense and Sensibility, and Secrets & Lies, indicating a strong preference for high-quality content. However, they also express dissatisfaction with lower-rated items, suggesting they are discerning viewers who appreciate quality content, particularly in genres that align with their tastes.
- Manager: Finish with enhanced answer:
 - The answer you give (may be INCORRECT). 1. USER ANALYSIS: User 123 has a diverse interaction history with a mix of ratings. They have rated several items highly, particularly enjoying Fantasia, Dead Man Walking, Sense and Sensibility, and Secrets & Lies, indicating a strong preference for high-quality content. However, they also express dissatisfaction with lower-rated items, suggesting they are discerning viewers who appreciate quality content, particularly in genres that align with their tastes. 2. MOVIE ANALYSIS: Beverly Hills Ninja (1997) - Action/Comedy
 - Summary: A bumbling but good-hearted man, raised by ninjas, embarks on a mission to prove himself as a true ninja while trying to rescue a kidnapped heiress. His clumsy antics lead to comedic situations as he learns the true meaning of bravery and honor.
 - Reason: User 123 has shown a preference for high-quality content, particularly enjoying movies with strong narratives and character development. While Beverly Hills Ninja offers humor and action, it may not align perfectly with the user's taste for more serious or dramatic films, hence the moderate predicted rating.
 - 3. PREDICTED RATING: Score: 3.5/5 stars.

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FUTURE DEVELOPMENT

- The system can be extended to a multimodal environment (text, image, and video), enabling more accurate recommendations for content with rich visual and auditory features.
- It can be further developed into a Recommendation-as-a-Service (RaaS) framework with standardized APIs, facilitating seamless integration into websites, mobile applications, and enterprise chatbots.
- Additionally, the system continuously monitors and exploits real-time data streams to adapt its models dynamically, thereby delivering timely, accurate, and user-adaptive recommendations.

CONCLUSION

- Successfully develops a movie recommendation system based on a Multi-Agent System architecture consisting of five agents: the Interpreter Agent, Manager Agent, Analyst Agent, Searcher Agent, and Reflector Agent.
- Each agent is assigned a distinct functional role and specialized responsibilities, collaboratively operating within the system to generate recommendations that satisfy user requirements.
- The system can be further refined and deployed in real-world recommender applications.